Higher Criticism for Adaptive Retrieval: From Pipeline Integration to a Controlled Study of Null Calibration

Anonymous submission

Abstract

Data retrieval has become a load-bearing component of modern AI systems: retrieving too few passages leaves a reader to hallucinate or miss the answer, while retrieving too many wastes tokens, inflates latency, and distracts the model with irrelevant context. We view retrieval as a sparse signal detection and identification problem: across a vast corpus, only a few entries, if any, are relevant to a given query, and their similarity scores must be separated from a dense background of near-misses. We therefore propose to select the retrieved set by a method based on Higher Criticism Thresholding (HCT), a well-established selection tool from the sparse signal detection literature. We apply our HCT-based method to cutoff document relevance scores in four Retrieval-Augmented Generation (RAG) pipelines. On multi-hop QA (HotpotQA), HCT raises context precision more than tenfold over gap-based Adaptive-K and improves answer Exact Match by 23% (relative), while reading an order of magnitude fewer tokens. These integrations inherit each pipeline's own null. The sharper test is whether HCT survives the single global null a deployed retriever must actually rely on: one calibration for every query. We put HCT through that test on a controlled corpus of 80,000 products with construction-time ground truth for every document. The result is informative: under a naive shared null HCT is overly conservative and returns nothing on more than 40% of queries. The collapse turns out to be not a weakness of the statistic but a scale mismatch between the shared null and each query's own score distribution, correctable by a label-free per-query rescaling that reads only observable pool statistics at inference time, with no additional data, no retraining, and no tuning. With the correction, HCT attains the best retrieval F1 of eight cutoff rules on this corpus (0.209 vs. 0.187 for the strongest fixed Top-k we evaluated, a 12% relative gain), and lifts end-to-end answer token-F1 by 15% over a Top-20 baseline on a 100,000-document electronics corpus with `gpt-5-mini`, yielding a query-adaptive cutoff that generalizes from hundreds to 100,000 documents without labeled retrieval data.

1 Introduction

Retrieval-Augmented Generation (RAG) (Lewis et al. 2020) grounds a large language model (LLM) in passages pulled from an external corpus by an approximate-nearest-neighbour search. The dominant interface is *Top-k*: fix an integer k and pass the k highest-scoring passages to the generator. This is convenient but blunt. A single k cannot serve a fact-lookup query that needs one passage, a multi-hop question that needs a handful, and a database-style aggregation that needs dozens. Because the number of genuinely relevant passages varies enormously across queries, one fixed k is wrong for most queries at once: too small a value discards evidence, while too large a value floods the prompt with distractors, inflating cost and degrading answers. What the retriever needs is not a better fixed k but a per-query rule that decides, from the scores themselves, where the relevant passages stop.

1.1 Retrieval as Sparse Signal Detection

A query-aware retriever maps every document in the corpus to a similarity score against the query. The overwhelming majority of those scores come from *irrelevant* passages and together form a background null distribution; only a small, query-dependent number of relevant passages (the *signals*) are pushed above it. Choosing k is then a *detection* problem rather than a ranking one: how many of the top scores are too high to be explained by the irrelevant background? This is the rare/weak-signal mixture problem that Higher Criticism was designed for, with documents in place of features and an empirical similarity null in place of an analytic Gaussian null.

1.2 Higher Criticism as a Cutoff Rule

Higher Criticism (HC) (Donoho and Jin 2004) is a goodness-of-fit statistic for detecting a small number of weak signals in a sea of nulls. Given a pool of N candidate documents with similarity scores $s_1 \geq \cdots \geq s_N$, we convert each s_i to a p-value p_i under a calibrated null. The p-value of a (query, document) pair is the probability of observing a similarity at least as high as s_i between the query and a randomly drawn *irrelevant* passage, so a small p_i means the document is too similar to be a noise hit. Sorting $p_{(1)} \leq \cdots \leq p_{(N)}$, the HC statistic at rank i is

$$\mathrm{HC}_i \;=\; \sqrt{N}\,\frac{i/N - p_{(i)}}{\sqrt{p_{(i)}(1 - p_{(i)})}},$$

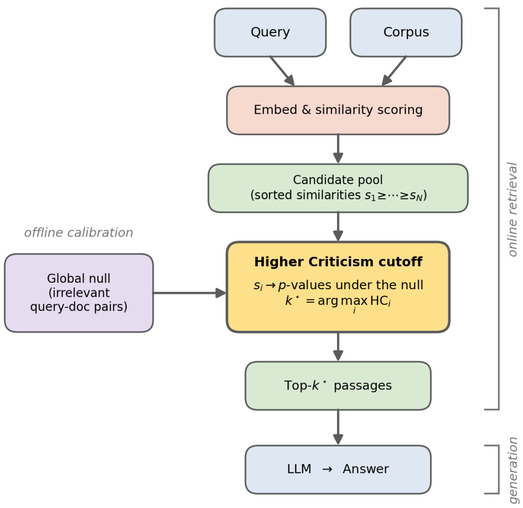

Figure 1: Overview of the HC-based retrieval cutoff. *Offline*, a single *global null* is calibrated from the similarity scores of irrelevant query–document pairs. *Online*, a query and corpus yield a candidate pool of sorted similarities; the Higher Criticism cutoff maps these scores to p-values under the null and selects the $k^\star = \arg\max_i \mathrm{HC}_i$ top passages, which the generator then reads. Only the cutoff box changes across the pipelines we evaluate.

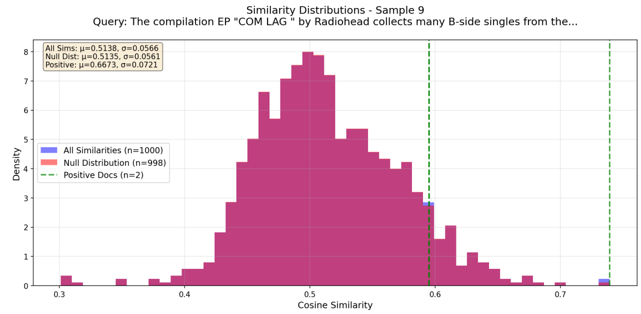

Figure 2: Retrieval as sparse signal detection (one HotpotQA query). The few relevant documents (green) sit in the right tail of a large null of irrelevant similarity scores; HC locates where that tail stops being explainable by the null.

which compares the empirical order-statistic quantile $p_{(i)}$ with its expectation i/N under the null, where the p-values would be i.i.d. uniform. A large positive HC_i means the top-i scores are systematically more extreme than the null predicts, and the adaptive cutoff is $k^\star = \arg\max_i \mathrm{HC}_i$. We bound the search to the top γ fraction of the pool only as a guardrail: at the very smallest ranks the denominator $\sqrt{p_{(i)}(1-p_{(i)})}$ is tiny, so a single outlier can create a spurious maximum. The γ window simply excludes that unstable tail and does not otherwise shape the selection.

1.3 The Null Distribution

Every quantity above is anchored to one object: the null distribution of similarity scores for irrelevant query–document pairs. It is the reference against which an observed score is judged extreme. We build it once, offline, by sampling many random pairings of a query with documents that are not relevant to it and standardizing the resulting similarities to z-scores, so that a single calibrated distribution summarizes what a typical irrelevant match looks like. At inference, an observed score s_i is mapped to its p-value as the null's tail mass above s_i, and those p-values feed the HC statistic. Because the p-values, the i/N comparison, the location of the maximum, and hence $k^\star$ all derive from this one mapping, the null is the single object that determines the entire cutoff. A retriever meant for deployment calibrates it once and must apply it to queries it has never seen, so the production-realistic requirement is a single *global* null shared by all queries. Per-query nulls are a controlled-lab convenience, not a deployable design, and the global null is the realistic constraint under which HC must operate.

1.4 Research Question

Our objective is not merely to benchmark HC but to understand *when* an HC-driven cutoff helps and to characterize and repair the cases where it does not, asking how the global null fails when its conditions break and whether those failures can be diagnosed from observable pool statistics and corrected without per-query labels. A second question, motivated by an early-stage observation that HC behaved more conservatively (higher precision, lower recall) than its theory predicts, is a direct check on the calibration itself: are the p-values under the null actually uniform, and if not, where does the deviation come from? Section 4.3 answers the first question in a controlled setting, and the uniformity diagnostic is verified in the supplement.

1.5 Our Contributions

1. **HC pipeline integrations.** We replace fixed Top-k with an HC-determined cutoff and integrate it, with no architectural change, into three established pipelines: the Adaptive-RAG router (Jeong et al. 2024); the dense long-context Adaptive-K setting (Taguchi, Maekawa, and Bhutani 2025), hosted in FlashRAG (Jin et al. 2025); and an end-to-end pipeline judged by an LLM on financial question answering (Thakur et al. 2021).
2. **CrossEntityQA, a new sparse-relevance benchmark.** A purpose-built multi-entity retrieval benchmark from Wikidata and Wikipedia, with a controllable and widely varying number of relevant passages per query (K=2–20) and only about 2% relevance, designed to stress-test adaptive-k methods in the regime HC targets (construction details in the supplement).
3. **Head-to-head evaluation, with a diagnosed and repaired failure.** Across five datasets and a panel of fixed and adaptive baselines we report retrieval and generation quality together with cost. In a controlled product-

retrieval setting this evaluation pinpoints *how* a single global null makes HC collapse on hard queries, and a label-free per-query recalibration recovers it, giving HC the best retrieval F1 of the methods compared while preserving one global null.

2 Related Work

Adaptive retrieval cutoffs. The default RAG interface (Lewis et al. 2020) passes a fixed Top-k set to the generator regardless of the score distribution. Several methods instead set the cutoff from the retrieved scores themselves. Gap-based Adaptive-K (Taguchi, Maekawa, and Bhutani 2025) cuts at the largest gap between consecutive sorted scores; the clustering rule CAR (Xu et al. 2025) places the boundary at a cluster edge; and Adaptive-RAG (Jeong et al. 2024) routes each query to a retrieval strategy by a predicted complexity class. These geometric and routing rules read the shape of the score curve or a query-level classifier rather than testing the scores against a calibrated background of irrelevant matches, which is the gap our detection view targets.

Multiple testing and goodness-of-fit. Casting the cutoff as deciding which top scores are too extreme to have come from an irrelevant-score null connects retrieval to multiple hypothesis testing. Bonferroni (Dunn 1961) controls the family-wise error rate and Benjamini–Hochberg (Benjamini and Hochberg 1995) the false discovery rate by thresholding individual p-values, whereas Berk–Jones (Berk and Jones 1979) and Higher Criticism (Donoho and Jin 2004) aggregate the whole ordered set of p-values into a goodness-of-fit deviation from uniformity. Higher Criticism was introduced for detecting a few weak signals sparsely mixed into a large null (Donoho and Jin 2004), the regime we argue matches sparse-relevance retrieval; we adopt it as a per-query cutoff and compare it against these alternatives on a shared p-value construction.

Retrievers, frameworks, and benchmarks. Our dense pipelines build on BGE embeddings (Xiao et al. 2023) indexed for approximate-nearest-neighbour search with FAISS (Johnson, Douze, and Jégou 2019), and register the HC cutoff as a post-processing stage in the FlashRAG toolkit (Jin et al. 2025). We evaluate across established retrieval and QA benchmarks—BEIR and its FiQA financial-QA task (Thakur et al. 2021), the multi-hop HotpotQA (Yang et al. 2018) and MuSiQue (Trivedi et al. 2022), and the long-context aggregation benchmark HoloBench (Maekawa, Iso, and Bhutani 2024)—and, to probe the sparse, variable-K regime directly, our own CrossEntityQA benchmark (construction in the supplement).

3 Methods

3.1 HC Retrieval Algorithm

The HC cutoff has two parts: an offline calibration of the null and an online selection rule. Offline, we sample random (query, non-relevant passage) pairs, standardise their similarity scores to z-scores, and fit the empirical CDF of that pooled distribution. This single calibrated null, shared by every query at inference, is the deployable object: a production retriever calibrates once and must then generalise to unseen queries, so we hold the null fixed rather than re-estimating it per query. Online, for a query we retrieve a candidate pool, map each score to a z-score, and pass it through the null's empirical CDF to obtain a p-value. Sorting the p-values, we take the cutoff $k^\star = \arg\max_i \mathrm{HC}_i$ (with the γ guardrail of Section 1.1) and return the top-$k^\star$ passages. Because the null governs every p-value, its fidelity to the current query's irrelevant scores is what decides whether the cutoff is well placed; Section 4.3 traces essentially all of HC's failures to this object and repairs them without abandoning the global null. The cutoff is a drop-in stage, so every pipeline we evaluate differs only in this box; the full offline/online pipeline is summarized in Figure 1. Algorithm 1 states the procedure step by step.

Algorithm 1 HCT-Ret: Higher-Criticism retrieval cutoff

Offline (once): sample random (query, non-relevant passage) pairs; standardise their similarities to z-scores; fit the empirical CDF $\widehat{F}_0$ of this global null.
Input: query q, candidate-pool size N
Parameter: guardrail fraction γ
Output: selected passages

retrieve the top-N passages for q with scores $s_1 \geq \cdots \geq s_N$
map each score to a z-score z_i
(optional, deployed setting) $z_i \leftarrow z_i \cdot \sigma_{\mathrm{null}}/\hat{\sigma}_q$ with $\hat{\sigma}_q = \mathrm{std}\big(z_{(\gamma+1:N)}\big)$ {label-free per-query recalibration (Sec. 4.3)}
$p_i \leftarrow 1 - \widehat{F}_0(z_i)$ {tail mass under the null}
sort $p_{(1)} \leq \cdots \leq p_{(N)}$
for $i = 1$ to $\lfloor \gamma N \rfloor$ **do**
  $\mathrm{HC}_i \leftarrow \sqrt{N}\,(i/N - p_{(i)})/\sqrt{p_{(i)}(1-p_{(i)})}$
end for
$k^\star \leftarrow \arg\max_{i \leq \gamma N} \mathrm{HC}_i$
return the top-$k^\star$ passages

3.2 Methods and Pipelines

We treat HC as one selection rule among a family that turns a candidate pool into a retrieved set, spanning three design philosophies. The *rank-based* rule is fixed Top-k, which ignores the score distribution. The *geometric* rules cut on the shape of the sorted scores: gap-based Adaptive-K (Taguchi, Maekawa, and Bhutani 2025) cuts at the largest score gap and clustering-based CAR (Xu et al. 2025) at a cluster boundary, while Adaptive-RAG (Jeong et al. 2024) routes each query by predicted complexity. The *statistical* rules operate on the same p-values as HC: Bonferroni (Dunn 1961) and Benjamini–Hochberg (Benjamini and Hochberg 1995) threshold individual p-values to control the family-wise error and false discovery rates, while Berk–Jones (Berk and Jones 1979) and HC (Donoho and Jin 2004) aggregate the whole ordered set into a goodness-of-fit deviation from the uniform null. Holding the pool and the p-value machinery fixed isolates the selection rule as the axis of comparison; the full method-by-method table is in the supplement.

Dataset	Docs	Queries	Relevant / query
Phase 1: HC inside three existing pipelines			
CrossEntityQA (ours)	13K	105	variable, 2–20
MuSiQue	139K	500	1–4
HotpotQA	2K	987	∼2
HoloBench	260	90	∼152
BEIR FiQA	58K	648	scattered
Phase 2: controlled study on our own pipeline			
City benchmark	726	44	variable, 4–24
Category+price products	80K	135	∼4.8%
Electronics products	100K	44	∼1.2%

Table 1: The evaluation suite. Phase 1 stresses an HC cutoff inside three pipelines HC did not control (the retriever for each is named in Tables 2–4); Phase 2 isolates mechanism on a pipeline with fully known relevance, moving from a benchmark where embeddings separate cleanly to product queries whose price constraint they cannot encode.

4 Experiments

4.1 Datasets and Setup

The study runs in two phases that serve different purposes (Table 1). Phase 1 tests HC inside three existing pipelines it did not control and across standard benchmarks; these fix the retriever and answerer, so they measure the end-to-end value of an adaptive cutoff but do not let us instrument the null. Phase 2 isolates mechanism on our own pipeline, where every document's relevance and every query's true number of relevant passages are fixed by construction, so each outcome can be attributed to a specific cause. Phase 2 is organised around one design principle: each benchmark is built so that a dense embedder cannot separate relevant from non-relevant documents by semantics alone, which keeps the task from being solved by plain nearest-neighbour ranking and genuinely exercises HC's statistical selection. It starts with a controlled *city-retrieval benchmark* of "largest cities in [country]" queries, where embeddings do separate relevant from irrelevant cleanly and a correctly calibrated cutoff should work. It then moves to product corpora whose queries pair a category with a price constraint (for example, "electronics under $150"): price is an attribute the text embedding does not encode, so items in the right category at the wrong price sit among the relevant ones and cannot be ranked apart by similarity. We report retrieval Recall/Precision/F1 and the mean adaptive k, and, where a generator is in the loop, answer EM and token-F1.

4.2 Integrating HC across Retrieval Pipelines

HC supplies a cutoff rule, not a retriever, so the first question is whether it transfers across the heterogeneous parts a real RAG system is built from: sparse and dense retrievers, routing classifiers, long-context readers, and LLM judges. We integrate an HC cutoff into three pipelines and measure what changes, interpreting the results pipeline by pipeline below (Tables 2–4). Each pipeline reports the metrics that matter for it. Throughout, we adopt each pipeline's stored null as given, because a single global null is a calibration choice we introduce only in Phase 2.

Strategy	Pass. F1	EM	Pass. P
CrossEntityQA (K=2–20)			
ONER (k=15)	0.609	0.152	0.671
IRCOT	0.540	0.105	0.697
HC	**0.620**	0.143	0.695
MuSiQue			
ONER (k=15)	0.301	0.164	—
IRCOT	0.315	0.204	—
HC	0.301	0.172	—

Table 2: Pipeline 1: Adaptive-RAG routing (BM25, `gpt-4o-mini`). Passage-retrieval F1, answer EM, and passage precision. HC wins F1 on the variable-K benchmark and ties single-step ONER on MuSiQue.

Method	Quality	Ctx P	Tokens
HotpotQA (answer F1, EM)			
Adaptive-K	0.306 / 0.122	0.020	25.3k
HC	**0.332 / 0.150**	**0.221**	**1.3k**
HoloBench (LLM-judge)			
Adaptive-K	0.326	0.531	20.2k
HC	0.321	**1.000**	**5.8k**

Table 3: Pipeline 2: Adaptive-K dense long-context (BGE-large, `gpt-4o-mini`). HC matches answer quality at far higher context precision and an order of magnitude fewer tokens. Quality is answer F1 / EM on HotpotQA and the LLM-judge score on HoloBench.

Where adaptivity pays: sparse, variable-K retrieval The clearest win is on CrossEntityQA, a sparse benchmark we built (construction in the supplement) so that the number of relevant passages swings from 2 to 20 per query. When the true count varies this widely, any fixed budget is wrong for most queries, and HC's adaptivity converts directly into retrieval quality: it leads on passage F1 by raising recall without giving up precision, where a fixed single-step retriever must trade one for the other. MuSiQue is the control. Each of its questions needs a small and fairly constant handful of passages, so adaptivity has little to exploit; HC neither helps nor hurts, matching single-step retrieval and trailing the more expensive iterative router only slightly. The contrast is the point: HC's benefit tracks how much the right k actually moves from query to query.

Same answer quality, far less context On the dense long-context pipeline, HC delivers the operating point an adaptive cutoff is supposed to produce: comparable answers from far less, cleaner context. On HotpotQA it gives the best answer EM and F1 in the comparison, and it does so while keeping about eleven of a thousand candidate chunks, which raises context precision by an order of magnitude and cuts input tokens roughly nineteen-fold against gap-based Adaptive-K.

Method	NDCG	Ret. P	Judge win
Baseline (k=5)	0.362	0.171	34.7%
HC	**0.418**	**0.319**	**44.8%**

Table 4: Pipeline 3: LLM-embeddings + `gpt-4o` judge on BEIR FiQA. Token-overlap answer quality is flat (omitted), but HC leads on ranking (NDCG) and retrieval precision, and the judge prefers its answers (44.8% of queries vs. 34.7%).

On HoloBench, an aggregation task where many passages are relevant, answer quality is a statistical tie, but HC reaches it with perfect context precision and several times fewer tokens and documents. In both cases the generation quality is matched or better while the retrieved set is smaller and cleaner, exactly the liberal-but-precise behaviour the theory predicts when the null is faithful.

When relevance is scattered: FiQA FiQA is the honest hard case. Its relevant passages are scattered and interleaved with the background rather than concentrated at the top, so there is no clean boundary for a detection rule to find, and token-overlap answer quality is flat across methods. Even here HC is not worse: it leads on ranking (NDCG) and retrieval precision, and once the γ guardrail is loosened enough to gather the dispersed passages, the `gpt-4o` judge prefers HC's answers on 44.8% of queries against 34.7% for the fixed baseline. The flat overlap score and the judge's preference are a metric divergence, not a contradiction. FiQA also gives us the p-value-uniformity diagnostic we return to in Phase 2: under an empirical null the non-relevant p-values are uniform while relevant documents carry systematically small ones, the signal HC is built to detect (see supplement).

4.3 A Controlled Study on Amazon

Phase 1 showed that the null distribution decides HC's behaviour, but the borrowed pipelines did not expose which queries failed or let us measure why. We therefore built our own environment: a family of product-retrieval datasets that fix every document's relevance and every query's true K by construction, with one controllable dimension of difficulty. With ground truth in hand, each outcome can be attributed to a mechanism. We read off the per-setting results in the tables below.

Proof of concept: the city benchmark The city benchmark is the regime built to satisfy HC's assumptions: 44 "largest cities in [country]" queries over 726 passages, with the number of relevant documents varying widely so adaptivity is necessary. Here BGE (Xiao et al. 2023) embeddings separate relevant from irrelevant cleanly, and a per-query empirical null is near-perfectly calibrated (pooled KS p=1.000). HC behaves as intended: its adaptive $k^\star$ tracks the true relevance count (Pearson r=0.835), and it beats every fixed cut on F1 by holding far higher precision at comparable recall (Table 5). This is the easy, well-calibrated baseline against which the harder corpora are deliberately contrasted.

A controlled difficulty: category-and-price queries To make HC's statistical selection necessary rather than optional, the product queries combine a category with a price constraint. A dense embedder captures the category but cannot reason about a numeric inequality, so items in the right category at the wrong price sit among the relevant ones in cosine space and no purely semantic ranking can separate them. Two corpora instantiate this: 80,000 products across three departments (135 queries, about 4.8% relevant) and a larger electronics corpus of 100,000 products (44 queries, about 1.2% relevant). Scaling to them also forces the calibration change that drives the rest of the study: we replace the city benchmark's per-query nulls with the single global null a deployable retriever must use.

Method	R	P	F1	k
Top-10 (best fixed)	0.800	0.766	0.783	10
HC	0.852	**0.938**	**0.893**	9.7

Table 5: Phase-2, city benchmark (726 docs; embeddings separate cleanly). HC beats the best fixed cut on F1 by holding far higher precision at comparable recall.

The failure. Under that global null, HC breaks on a large share of the queries. 55 **of** 135 **produce a negative HC statistic and return** k=0, retrieving nothing. Because relevance and true K are known, we can ask exactly which queries collapse and why.

Diagnosis: three mechanisms Partitioning the 135 queries by their known structure assigns each collapse to one of three mechanisms. First, an **embedding limitation** accounts for 24 "irreducible" queries: the price signal leaves no trace in the cosine geometry, the relevant documents are scattered through the ranks, and no test on these scores can recover a signal that is not there. Second, a **calibration mismatch** accounts for 31 "fixable" queries: the embedding does separate the relevant items, but a query's non-relevant scores are more tightly spread ($\sigma_q \approx 1.74$) than the shared null ($\sigma_{\text{null}} \approx 2.06$), which inflates the p-values and trips the negative-HC gate. This scale ratio is the dominant predictor of failure, correlating at r=0.83 with the HC statistic. Third, **signal dilution** is a general pressure: as the candidate pool N grows the win condition $p_{(i)} < i/N$ tightens, so a bigger pool is not a free fix. The important split is that the fixable queries fail from miscalibration, not missing signal, and the miscalibration is a direct consequence of sharing one global null across queries of different scale. The fix must therefore recalibrate per query without giving up the global null.

The fix and what it recovers The repair rescales each query's z-scores onto the null's scale before the lookup. We estimate a robust per-query spread $\hat{\sigma}_q$ from the pool after dropping the top γ ranks (where any relevant documents concentrate and would distort the estimate), then set

$$z'_i = z_i \cdot \frac{\sigma_{\text{null}}}{\hat{\sigma}_q}, \qquad \hat{\sigma}_q = \operatorname{std}\big(z_{(\gamma+1\,:\,N)}\big)\,.$$

The transform uses only observable pool statistics, so no labels enter, and it is applied uniformly, so the null itself stays one global distribution. It cancels the scale mismatch

Method	R	P	F1	k
Top-20	0.179	0.192	0.185	20
Top-50	0.287	0.139	0.187	50
Adaptive-K (gap)	0.104	0.204	0.138	9
CAR (cluster)	0.598	0.060	0.110	417
Bonferroni	0.014	0.049	0.022	0.3
Benjamini–Hochberg	0.064	0.087	0.074	1.6
Berk–Jones + recal	0.238	0.176	0.202	35
HC + recal	0.209	0.209	**0.209**	34

Table 6: Phase-2, category-and-price corpus (80K docs, pool N=1000). HC with recalibration is the only method whose F1 beats every fixed Top-k; the cluster rule over-retrieves (k=417) and the multiple-testing rules collapse to a handful of documents.

directly. The fix is targeted, and it only ever helps. The calibration-mismatch group, which the global null had zeroed out, returns to roughly the fixed-k baseline (macro-F1 $0.000 \to 0.157$, 30 of 31 queries recovered); the embedding-limited group stays near zero, because nothing in its scores can be recovered; and the 77 already-working queries are left untouched ($0.253 \to 0.251$). The dominant failure mode is removed without labels and without changing the null. The payoff shows across the whole field: once recalibrated, HC posts the best F1 of all eight cut-off methods (Figure 3).

Best in class, and it generalizes With the recalibration in place, HC is the only method whose F1 beats every fixed Top-k on the category-and-price corpus, at a moderate mean k=34 (Table 6). The contrast with the other adaptive rules is instructive. The clustering rule over-retrieves to hundreds of documents and the classical multiple-testing rules under-retrieve to almost none, so neither finds a useful operating point; only the goodness-of-fit statistics (HC, and Berk–Jones behind it) sit where a liberal-but-precise cutoff should. The advantage is not a small-corpus artifact. On the 100,000-document electronics corpus HC again takes the best retrieval

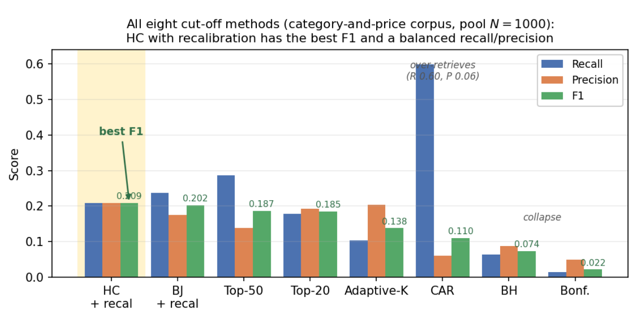

Figure 3: All eight cut-off methods on the category-and-price corpus (pool N=1000). HC with recalibration (highlighted) has the best F1 and a balanced recall/precision; the cluster rule (CAR) over-retrieves to a high recall but near-zero precision, and the family-wise-error (Bonferroni) and false-discovery (BH) rules collapse to almost nothing.

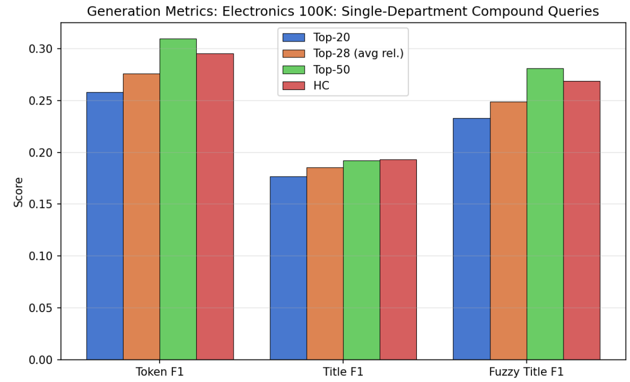

Figure 4: End-to-end generation on the 100K electronics corpus. HC reaches 94–97% of a fixed Top-50 cut's answer quality while retrieving about a third fewer documents, and improves on Top-20 across token-, title-, and fuzzy-title-F1.

F1 (0.192 vs. 0.185 and 0.166 for fixed Top-20/50), and the gain carries through generation: with `gpt-5-mini` it improves answer token-F1 by $+15.0\%$ over a fixed Top-20 and reaches 94–97% of a much larger Top-50 cut's quality while reading about a third fewer documents (Figure 4).

Uniformity of the p-values. The negative-HC collapse described above is the symptom of inflated p-values, the scale mismatch is its measured cause, and the recalibration restores the expected recall-up, precision-down behaviour. We also verify uniformity directly: under an empirical null the non-relevant p-values are uniform, whereas a parametric normal null is not; the diagnostic plots are in the supplement.

5 Conclusions and Future Work

Higher Criticism is a competitive query-adaptive cutoff whose behaviour is governed by the null distribution that turns similarities into p-values and fixes the threshold. That dependence is true by construction; what matters is a precise, measured account of how the null behaves at the boundary of detectability (Donoho and Jin 2004). In a controlled setting we showed how a single global null, the production-realistic requirement that a deployable retriever calibrate once and generalize to unseen queries, fails on hard queries through a per-query scale mismatch, and we corrected it with a label-free recalibration that needs no labels and keeps one global null. The recalibrated method gives the best retrieval F1 of any we tested (0.209, the only one to beat every fixed Top-k), and the advantage holds to 100,000 documents and through to generated answers.

The residual failures are embedding-limited, not statistical: when a query turns on a numeric attribute the embedding cannot encode, no test recovers an absent signal. The natural next step is hybrid retrieval that pairs embeddings for category semantics with structured metadata filtering for the numeric constraint, with a per-query null for small clustered corpora and a benchmark-wide p-value uniformity diagnostic as further directions.

6 Ethics Statement

This work uses only publicly available datasets under their published licences (BEIR, HotpotQA, MuSiQue, HoloBench, and public dumps of Wikidata and Wikipedia) and public product-catalogue snapshots. No personally identifiable data is collected, and no human-subjects experiment is conducted. The end-to-end evaluation calls the OpenAI API family (`gpt-4o`, `gpt-4o-mini`, `gpt-5-mini`) and uses these models both as generators and as automatic judges; we report this dual role transparently. Generative-AI writing tools were used in a limited copy-editing capacity for grammar and phrasing; they were not used to generate research design, experimental results, or analysis, and all reported numbers are produced by the described code. We do not foresee direct dual-use or malicious-use risks that are specific to this work beyond those already present in generic RAG systems.

References

Benjamini, Y.; and Hochberg, Y. 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, 57(1): 289–300.

Berk, R. H.; and Jones, D. H. 1979. Goodness-of-Fit Test Statistics that Dominate the Kolmogorov Statistics. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 47(1): 47–59.

Donoho, D.; and Jin, J. 2004. Higher Criticism for Detecting Sparse Heterogeneous Mixtures. *The Annals of Statistics*, 32(3): 962–994.

Dunn, O. J. 1961. Multiple Comparisons Among Means. *Journal of the American Statistical Association*, 56(293): 52–64.

Jeong, S.; Baek, J.; Cho, S.; Hwang, S. J.; and Park, J. C. 2024. Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity. In *Proceedings of NAACL*.

Jin, J.; Zhu, Y.; Yang, X.; Zhang, C.; and Dou, Z. 2025. FlashRAG: A Modular Toolkit for Efficient Retrieval-Augmented Generation Research. In *Proceedings of the ACM Web Conference (WWW)*.

Johnson, J.; Douze, M.; and Jégou, H. 2019. Billion-Scale Similarity Search with GPUs. *IEEE Transactions on Big Data*.

Lewis, P.; Perez, E.; Piktus, A.; Petroni, F.; Karpukhin, V.; Goyal, N.; Küttler, H.; Lewis, M.; Yih, W.-t.; Rocktäschel, T.; Riedel, S.; and Kiela, D. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Proceedings of NeurIPS*.

Maekawa, S.; Iso, H.; and Bhutani, N. 2024. Holistic Reasoning with Long-Context LMs: A Benchmark for Database Operations on Massive Textual Data. In *Proceedings of NeurIPS Datasets and Benchmarks*.

Taguchi, C.; Maekawa, S.; and Bhutani, N. 2025. Efficient Context Selection for Long-Context QA: No Tuning, No Iteration, Just Adaptive-k. In *Proceedings of EMNLP*. Megagon Labs.

Thakur, N.; Reimers, N.; Rücklé, A.; Srivastava, A.; and Gurevych, I. 2021. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. In *Proceedings of NeurIPS Datasets and Benchmarks*.

Trivedi, H.; Balasubramanian, N.; Khot, T.; and Sabharwal, A. 2022. MuSiQue: Multihop Questions via Single-hop Question Composition. *Transactions of the Association for Computational Linguistics*, 10: 539–554.

Xiao, S.; Liu, Z.; Zhang, P.; and Muennighoff, N. 2023. C-Pack: Packaged Resources to Advance General Chinese Embedding. https://huggingface.co/BAAI/bge-base-en-v1.5.

Xu, Y.; Gupta, V.; Aggarwal, R.; Mahadevan, V.; and Krishnamachari, B. 2025. Cluster-based Adaptive Retrieval: Dynamic Context Selection for RAG Applications. arXiv:2511.14769.

Yang, Z.; Qi, P.; Zhang, S.; Bengio, Y.; Cohen, W. W.; Salakhutdinov, R.; and Manning, C. D. 2018. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. In *Proceedings of EMNLP*.